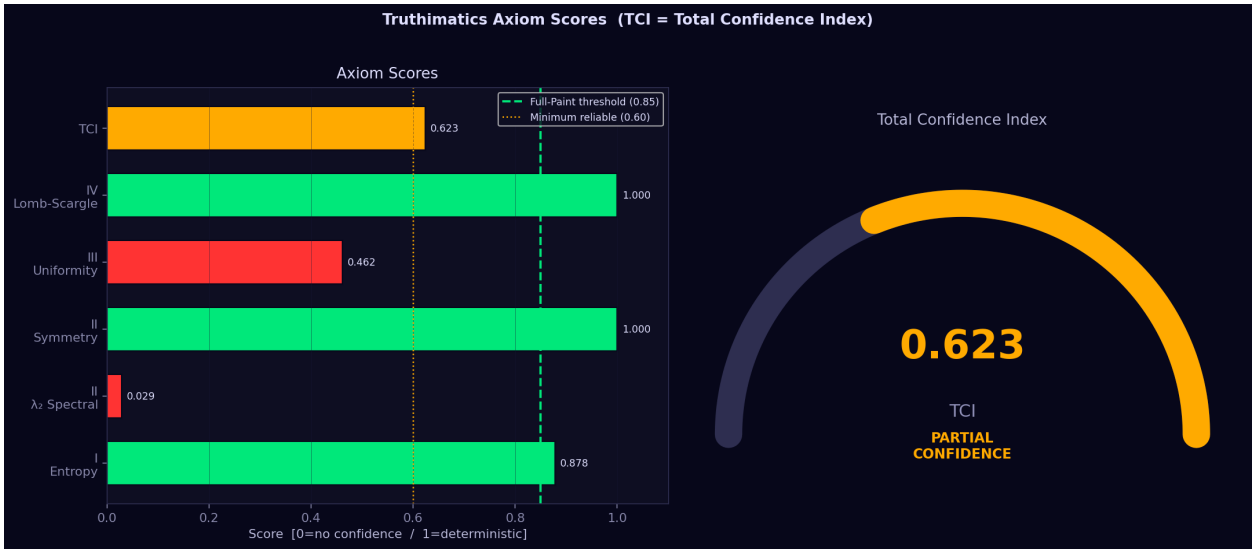
AXIOM-DINOSAUR v4.0

Truthimatics Full-Paint Deterministic Synthesis

Specimen ID	AXIOM_8F6F21C842
Taxa	Pteranodon
Clade	Pterosaur
PBDB Records	42
Mass (regression)	30.4 kg (Campione & Evans 2012)
Mass (volumetric)	13.1 kg (Silhouette integral, ρ=1060.0 kg/m^3)
Body Volume	0.017 m^3
Validity Score	1.0000 / 1.0000
TCI	0.6233 (partial)
Axiom I Entropy	0.8782
Axiom II λ■	0.1171 (symmetry 100%)
Axiom III Uniformity	0.4615
Axiom IV Lomb FAP	1.000000 peak 0.000 Hz
Integrity	66.7% Measured / 33.3% Inferred
Elements	24
CoG (m)	[0.086, 0.218, 0.0]
Generated	2026-05-01 00:03 UTC

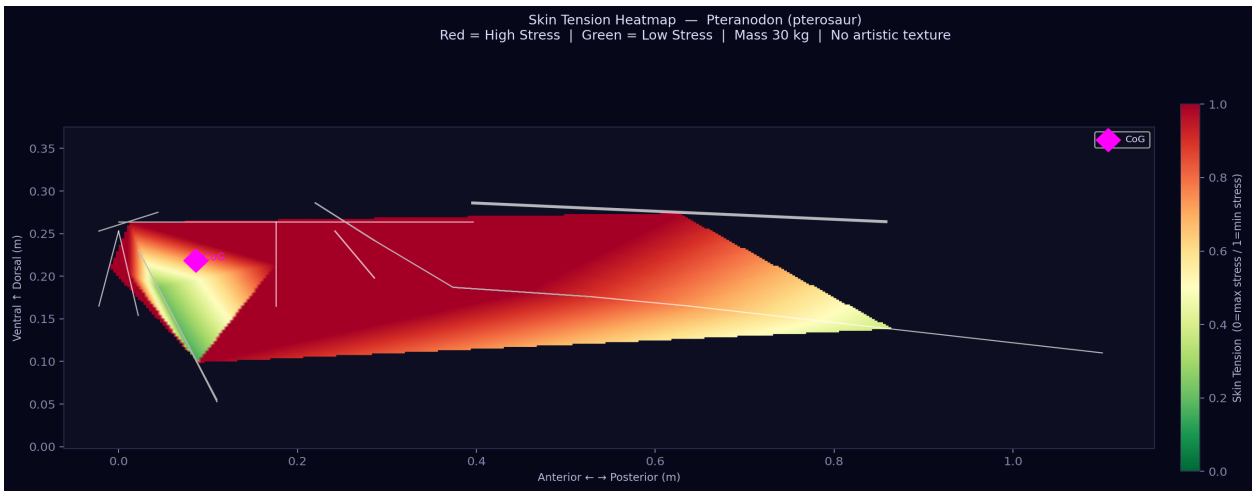
1 Truthimatics Confidence Analysis

TCI = weighted sum: Axiom I (entropy, w=0.25) + Axiom II (spectral, w=0.30) + Axiom III (Biewener uniformity, w=0.30) + Axiom IV (Lomb, w=0.15). TCI ≥ 0.85 authorises Full-Paint output.



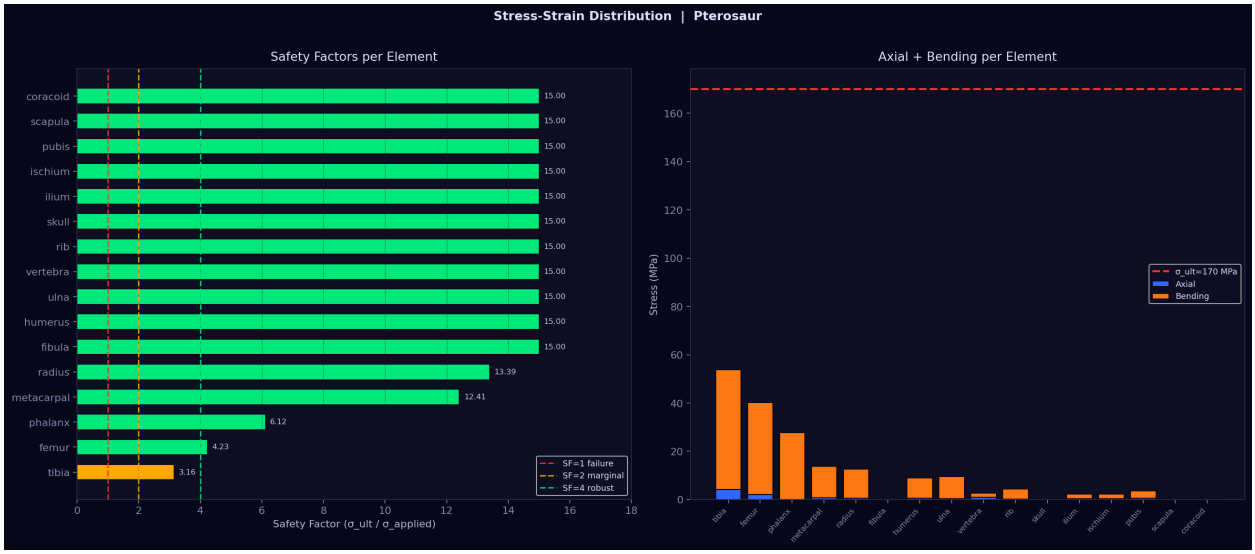
2 Skin Tension Heatmap

Stress field interpolated from per-bone safety factors onto the 2D body silhouette. Red=high stress (low SF), Green=low stress (high SF). No artistic texture applied.



3 Stress-Strain Distribution

Loading: biped (F=Mg). σ_ult=170 MPa (Reilly & Burstein 1975). SF≥4 robust.

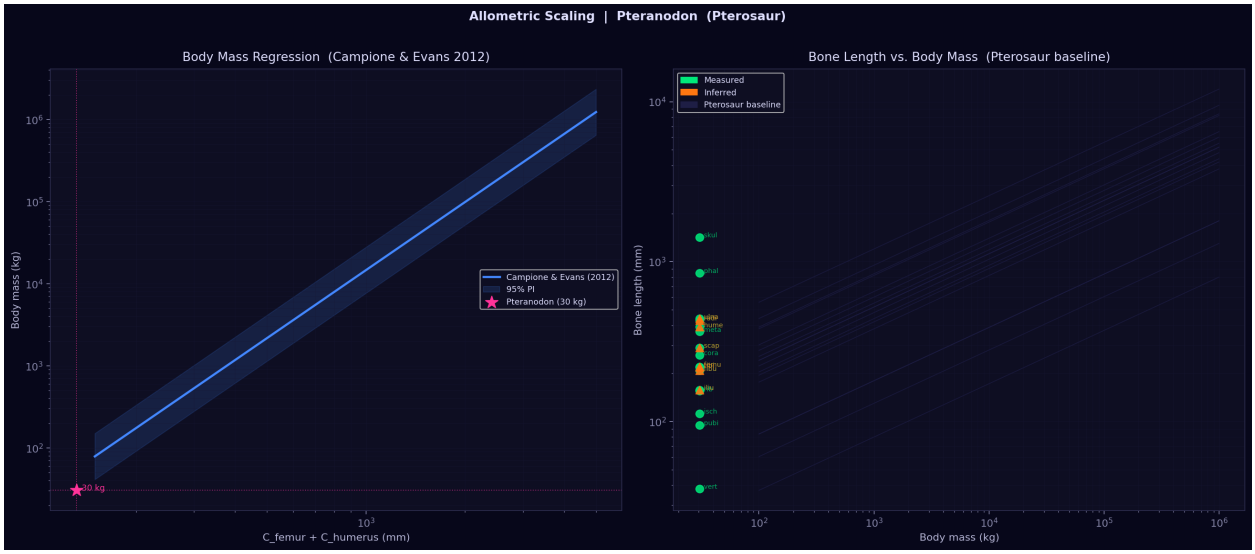


Per-Element Stress Summary

Element	σ_ax MPa	σ_bend MPa	σ_total MPa	SF	Status
coracoid	0.026	0.102	0.128	15.000	OK
femur	2.147	38.025	40.172	4.232	OK
fibula	0.000	0.000	0.000	15.000	OK
humerus	0.537	8.426	8.963	15.000	OK
ilium	0.411	1.829	2.240	15.000	OK
ischium	0.368	1.859	2.227	15.000	OK
metacarpal	0.799	12.896	13.696	12.413	OK
phalanx	0.000	27.784	27.784	6.119	OK
pubis	0.575	3.080	3.655	15.000	OK
radius	0.455	12.239	12.694	13.392	OK
rib	0.143	4.286	4.430	15.000	OK
scapula	0.028	0.129	0.157	15.000	OK
skull	0.000	0.000	0.000	15.000	OK
tibia	4.208	49.628	53.836	3.158	WARNING
ulna	0.368	9.130	9.498	15.000	OK
vertebra	0.869	1.693	2.562	15.000	OK

4 Allometric Scaling Analysis

Campione & Evans (2012): log■■(M_g)=2.754×log■■(C_f+C_h)–1.097 (R^2=0.97; N=426 non-avian dinosaurs).



5 Physical Properties Analysis

Biomechanical performance estimates derived deterministically from skeletal geometry. Speed: Alexander (1976) locomotion equation. Bite force: Christiansen & Wroe (2007). All values represent physically constrained estimates, not artistic approximations.

Parameter	Value	Notes
Hip Height	0.515 m	Femur + tibia + metatarsal
Estimated Speed	4.1 km/h (1.14 m/s)	Gait: Walk | Fr=0.259
Stride Length	0.788 m	Alexander (1976) / Thulborn (1990)
Bite Force	1084.0 N (110.5 kgf)	Pterosaur regression (Witton 2010)
Skull Length	1420.0 mm	Mid-shaft proxy from skull feature vector
Femoral Robusticity	0.1999	Circumference / length (dimensionless)
Limb Strength Index	1.2e-05	(A_f + A_t) / M^0.67 (Christiansen 1998)
Est. Body Length	2.05 m	Allometric from femur length
Est. Neck Length	0.61 m	Allometric from femur length
Est. Tail Length	0.16 m	Allometric from femur length

6 Complete Bone Manifest

Element	L (mm)	D (mm)	Vol cm³	Mass g	Method	Conf	SF
coracoid	260.0	75.0	585.81	731.3	Measured	0.88	15.00
femur	220.0	14.0	17.27	21.6	Measured	0.88	4.23
femur_right	220.0	14.0	17.27	21.6	Symmetry	0.84	4.23
fibula	208.0	4.0	0.94	1.0	Measured	0.88	15.00
fibula_right	208.0	4.0	0.94	1.0	Symmetry	0.84	15.00
humerus	390.0	28.0	122.47	152.9	Measured	0.88	15.00
humerus_right	390.0	28.0	122.47	152.9	Symmetry	0.84	15.00
ilium	158.0	32.0	64.81	80.9	Measured	0.88	15.00
ilium_right	158.0	32.0	64.81	80.9	Symmetry	0.84	15.00
ischium	112.0	20.0	17.94	22.4	Measured	0.88	15.00
metacarpal	365.0	25.0	34.04	28.0	Measured	0.88	12.41
phalanx	850.0	16.0	26.25	20.3	Measured	0.88	6.12
pubis	95.0	16.0	9.74	12.2	Measured	0.88	15.00
radius	430.0	18.0	55.81	69.7	Measured	0.88	13.39
radius_right	430.0	18.0	55.81	69.7	Symmetry	0.84	13.39
rib	155.0	7.0	3.04	3.8	Measured	0.88	15.00
scapula	290.0	72.0	602.18	751.7	Measured	0.88	15.00
scapula_right	290.0	72.0	602.18	751.7	Symmetry	0.84	15.00
skull	1420.0	155.0	13665.07	17058.1	Measured	0.88	15.00
tibia	218.0	10.0	8.73	10.9	Measured	0.88	3.16
tibia_right	218.0	10.0	8.73	10.9	Symmetry	0.84	3.16
ulna	440.0	20.0	70.50	88.0	Measured	0.88	15.00
ulna_right	440.0	20.0	70.50	88.0	Symmetry	0.84	15.00
vertebra	38.0	22.0	7.37	9.2	Measured	0.88	15.00